

ACADEMIC DETAILS				
Degree	Specialization	Institute	CPI/%	Year
B.Tech.	Computer Science & Engineering	IIT Gandhinagar	8.80	2024–Present
Class XII	Physics, Chemistry, Maths	Modern Vidya Niketan, Faridabad	95.4	2022–2024
Class X		Modern Vidya Niketan, Faridabad	96.8	2015–2022

- RELEVANT COURSEWORK
- **Computer Science:** Data Structures and Algorithms, Theory of Computing, Discrete Mathematics
 - **Artificial Intelligence:** Probability, Statistics & Data Visualization, Machine Learning, Introduction to Data Science
 - **Mathematics:** Calculus of Several Variables & Linear Algebra, Ordinary Differential Equations
 - **Finance:** Financial Modelling and Engineering

- EXPERIENCES
- **Under Vehicle Surveillance System | Project Researcher, CVIG Lab, IIT Gandhinagar** [Aug '25 – Nov '25]
Advisor: Prof. Shanmuganathan Raman | C++, Python, OpenCV, OpenPano | [Live Demo](#)
 - Calibrated a fisheye and planar cameras using Scaramuzza’s OCamCalib model, enabling image undistortion for panoramic vehicle undercarriage imaging.
 - Developed an end-to-end computer vision pipeline for automated contraband detection, using OpenCV and OpenPano to capture, undistort, and stitch vehicle undercarriage images with SIFT and RANSAC; deployed the pipeline through a Streamlit interface.

- PROJECTS
- **Fairness in Federated Learning: Client-Side Interventions, Server Protocols & Adaptive Weighting** [Jan '26 – Apr '26]
Advisor: Prof. Manisha Padala | Federated Learning, CNNs | [GitHub](#)
 - Developed a federated learning framework on UTKFace (23k images) and FairFace (50k images) datasets for gender classification across 10 clients over 8 rounds; established baseline race-wise fairness gaps of 8.52% (UTKFace) and 9.92% (FairFace).
 - Proposed **Dynamic Client-Adaptive Fairness Weighting (DCA-FW)**: designated fairness clients reweight per-group cross-entropy loss each round, reducing the UTKFace gap 8.52% → 5.07% while maintaining ~87% accuracy and validated on FairFace (gap: 9.92% → 8.12%).
 - Benchmarked four fairness interventions (static weighting, fairness regularization, adversarial fairness via race-adversarial training, attack-based) using SPD and EOD metrics; studied label-flipping attacks demonstrating the disproportionate harm to minority groups under FedAvg.
 - **Semantic Search Engine: Dense Retrieval & Hybrid Ranking** [Jan '26 – Apr '26]
Advisor: Prof. Anirban Das Gupta | BERT, FAISS, HNSW, BM25 | [GitHub](#) | [Live Demo](#)
 - Built a semantic retrieval engine over a 50K-document MS MARCO corpus using BERT embeddings, BM25, FAISS vector indexing, and hybrid ranking; implemented exact and HNSW-based ANN search, achieving 95.7% Recall@10 while significantly reducing retrieval latency.
 - Evaluated the system on 200 MS MARCO queries, where BERT-based semantic retrieval improved Precision@10 and Recall@10 by ~18% over BM25 and achieved nearly 10× higher effectiveness than Word2Vec-based retrieval.
 - **Algorithmic Trading Strategy Backtesting Platform** [Jan '26 – Apr '26]
Advisor: Prof. Mithun Radhakrishnan | Python, yfinance, Pandas, Streamlit | [GitHub](#) | [Live Demo](#)
 - Built a modular backtesting framework implementing four strategies — Moving Average Crossover, RSI Momentum, Mean Reversion (Z-score), and Breakout — over Yahoo Finance historical data with configurable stock and timeframe selection.
 - Developed a Streamlit dashboard with multi-strategy comparison, cumulative return curves, and performance metrics (total return, Sharpe Ratio, max drawdown) in interactive chart and table form.
 - **RAG Based AI Teaching Assistant** [Dec '25]
Python, Whisper, Embeddings | [GitHub](#) | [Live Demo](#)
 - Built an end-to-end RAG pipeline converting lecture videos into a searchable knowledge base: MP4 → FFmpeg audio → Whisper transcription → overlapping chunk embeddings → cosine-similarity retrieval → LLM-grounded generation.
 - Integrated OpenAI GPT and locally-run DeepSeek for context-aware generation; joblib backed embedding store enables incremental ingestion without full recomputation, designed to scale to FAISS / vector DB for production.

TECHNICAL SKILLS

- **Languages:** Python, C/C++
- **ML Frameworks:** PyTorch, TensorFlow, scikit-learn
- **Libraries:** OpenCV, FAISS, NumPy, Pandas, Matplotlib, Streamlit, yfinance
- **Tools & Technologies:** Git, GitHub, Docker

ACHIEVEMENTS

- **Codeforces: Specialist** (Max Rating: 1400+); solved **400+** DSA problems across LeetCode and Codeforces.
- Selected as a Mentee for the **Barclays Mentorship Programme 2026**, a competitive mentorship initiative providing exposure to investment banking, global markets, and the finance industry.
- Secured **AIR 1836** in JEE Advanced and **AIR 6362** (99.6 percentile) in JEE Mains, among **14 lakh+** candidates nationwide.
- Recipient of **100% Scholarship** through the **Shri Gopal Sharma Scholarship Test** for Classes XI–XII.
- Completed the **IIT Gandhinagar Explorer Fellowship**, a fully sponsored 42-day expedition across multiple Indian states.
- Received the **SOF Science Olympiad Gold Medal** (School level, Class IX).
- Won **Silver Medal** in IIT Gandhinagar Basketball League’2025.

POSITIONS OF RESPONSIBILITY

- **Transportation Coordinator, Welfare Council, IIT Gandhinagar** [Jun ’26 – Present]
 - Managing student transport logistics for the campus community, coordinating bus schedules, resolving operational issues, and improving accessibility for students.
- **Student Guide, IIT Gandhinagar** [May ’26 – Present]
 - Mentoring 10 first-year students from the 2026 batch in navigating academic, social, and campus life at IITGN; serving as primary point of contact through orientation and early-semester guidance.
- **Finance Member, Sports Council, IIT Gandhinagar** [Jun ’25 – May ’26]
 - Contributed to budgeting, event coordination, and operational planning for Sports Council events at IIT Gandhinagar.
- **Senior Marketing Executive, Blithchron (Cultural Fest), IIT Gandhinagar** [Apr ’25 – Apr ’26]
 - Led outreach, sponsorship acquisition, and digital marketing campaigns for Blithchron, IIT Gandhinagar’s annual cultural festival.